

CELLONA, BRYAN DION J-FA3

Github Link: <https://github.com/bryandioncellona/APM1110/blob/main/FA3/CELLONA%2C%20BRYAN%20DION%20J-FA3.Rmd>

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5. A geospatial analysis system has four sensors supplying images. The percentage of images supplied by each sensor and the percentage of images relevant to a query are shown in the following table.

```
# Given
sensor <- c(1, 2, 3, 4)
images_supplied <- c(15, 20, 25, 40)
relevant_images <- c(50, 60, 80, 85)

sensor_data <- data.frame(
  Sensor = sensor,
  Percentage_Images_Supplied = images_supplied,
  Percentage_Relevant_Images = relevant_images
)

sensor_data
```

##	Sensor	Percentage_Images_Supplied	Percentage_Relevant_Images
## 1	1	15	50
## 2	2	20	60
## 3	3	25	80
## 4	4	40	85

What is the overall percentage of relevant images?

```
# Compute
overall_relevant <- sum(images_supplied * relevant_images) / sum(images_supplied)

# Output the result
cat("The overall percentage of relevant images is ", overall_relevant, "%.")

## The overall percentage of relevant images is 73.5 %.
```

6. A fair coin is tossed twice. Let E_1 be the event that both tosses have the same outcome, that is $E_1 = (HH, TT)$. Let E_2 be the event that the first toss is a head, that is, $E_2 = (HH, HT)$. Let E_3 be the event that the second toss is a head, that is, $E_3 = (TH, HH)$. Show that E_1, E_2 , and E_3 are pairwise independent but not mutually independent.

```
# Define the sample space
sample <- c("HH", "HT", "TH", "TT")

# Define events
E1 <- c("HH", "TT")
E2 <- c("HH", "HT")
E3 <- c("HH", "TH")

# Calculate probabilities
P_E1 <- length(E1) / length(sample)
P_E2 <- length(E2) / length(sample)
P_E3 <- length(E3) / length(sample)

P_E1_E2 <- length(intersect(E1, E2)) / length(sample)
P_E1_E3 <- length(intersect(E1, E3)) / length(sample)
P_E2_E3 <- length(intersect(E2, E3)) / length(sample)

P_E1_E2_E3 <- length(Reduce(intersect, list(E1, E2, E3))) / length(sample)

# Check for pairwise independence
pairwise_independent <-
  (P_E1_E2 == P_E1 * P_E2) &
  (P_E1_E3 == P_E1 * P_E3) &
  (P_E2_E3 == P_E2 * P_E3)

# Check for mutual independence
mutual_independent <- pairwise_independent &
  (P_E1_E2_E3 == P_E1 * P_E2 * P_E3)

# Display results
cat("P(E1) =", P_E1, "\n")
```

```
## P(E1) = 0.5
```

```
cat("P(E2) =", P_E2, "\n")
```

```
## P(E2) = 0.5
```

```
cat("P(E3) =", P_E3, "\n")
```

```
## P(E3) = 0.5
```

```

cat("P(E1 and E2) =", P_E1_E2, "\n")

## P(E1 and E2) = 0.25

cat("P(E1 and E3) =", P_E1_E3, "\n")

## P(E1 and E3) = 0.25

cat("P(E2 and E3) =", P_E2_E3, "\n")

## P(E2 and E3) = 0.25

cat("P(E1 and E2 and E3) =", P_E1_E2_E3, "\n")

## P(E1 and E2 and E3) = 0.25

cat("Pairwise Independent:", pairwise_independent, "\n")

## Pairwise Independent: TRUE

cat("Mutually Independent:", mutual_independent, "\n")

## Mutually Independent: FALSE

cat("\nConclusion: The events E1, E2, and E3 are pairwise independent but not mutually independent.\n")

##
## Conclusion: The events E1, E2, and E3 are pairwise independent but not mutually independent.

```